Sharp disagreement between egalitarian core selections in assignment games

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Abstract

For weighted bipartite assignment games on $n \geq 4$ players, we determine the maximum normalized ℓ_1 distance between the full leximin and leximax-min core allocations:

$$s_n^{\text{assign}} = \frac{\lfloor n/2 \rfloor}{\lfloor n/2 \rfloor + 1}.$$

The main structural lemma states that leximin gives each player at least half their payoff in every other core allocation. Matching complementarity and a minimax directional witness yield the sharp finite bound. A complete market with identical rows and weights in $\{1, 2\}$ attains it. Exact assignment games instead have coincident selections, a consequence of known pairwise egalitarian structure. External correctness and novelty review remain pending.

1 Definitions and main result

Let $G = (L, R, E)$ be a finite bipartite graph with nonnegative real edge weights w. Its assignment game values a coalition by the maximum matching weight in its induced subgraph. Write $n = |L|+|R|$ and let $T > 0$ be the maximum matching weight in G. Its entire core is

$$C = \left\{ p \geq 0 : \sum_a p_a = T, \quad p_u + p_v \geq w_{uv} \text{ for every } uv \in E \right\}. \tag{1}$$

The edge inequalities imply all coalition inequalities by summing over a coalition's optimal matching; the reverse follows from two-player coalitions. Nonemptiness follows from bipartite matching LP duality.

Let $x = \mathrm{Lmin}(C)$ lexicographically maximize the increasingly sorted payoffs, and let $z = \mathrm{Lmax}(C)$ lexicographically minimize the decreasingly sorted payoffs. Retaining players whose core payoff is identically zero does not change the other coordinates of either full selection. Define

$$\rho(C) = \frac{\|x - z\|_1}{T}, \qquad s_n^{\text{assign}} = \sup_{G:\, |L|+|R|=n,\, T>0} \rho(C).$$

Isolated players are allowed, and n counts them. No exact-game hypothesis is imposed.

Theorem 1. *Fix any optimal matching, delete its zero-weight edges, and let $k \geq 1$ be its number of remaining edges. For full leximin x and* any *core allocation z minimizing the largest payoff,*

$$\frac{\|x - z\|_1}{T} \leq \frac{k}{k+1}.$$

For $k = 1$ the selections coincide. For every $k \geq 2$ the bound is attained by a complete $k \times k$ market with weights in $\{1, 2\}$. Hence $s_n^{\text{assign}} = 0$ for $n = 2, 3$, and $s_n^{\text{assign}} = \lfloor n/2 \rfloor / (\lfloor n/2 \rfloor + 1)$ for $n \geq 4$. In particular, $s_n^{\text{assign}} \to 1$.

The result applies to nonunique optimal matchings and minimax faces. Full leximin is essential; a first-stage maximin point need not suffice.

2 A simultaneous half-payoff guarantee

Fix the matching from Theorem 1, with positive edge weights $s_1, \ldots, s_k$. Summing its inequalities in (1) and using efficiency shows that every matched pair has total s_i, and every unmatched player receives zero throughout the core. These identities require no nondegeneracy or exactness assumptions.

Lemma 2. *For every $y \in C$ and every player a, $y_a \leq 2x_a$. Consequently $x_a \geq \frac{1}{2} \max_{y \in C} y_a$ simultaneously for all players, and $\|x - y\|_1 / T \leq 1$ for every $y \in C$.*

Proof. Put $d = y - x$. Suppose first that $a \in L$ and $d_a = \delta > 0$. Set

$$A = \{u \in L : d_u \geq \delta\}, \qquad B = \{v \in R : d_v \leq -\delta\}.$$

Matched-pair totals give opposite changes across each matching edge; all unmatched changes vanish. Thus the matching bijects A and B. At an edge uv tight at x, feasibility of y implies $d_u + d_v \geq 0$. Such an edge cannot have $u \notin A$ and $v \in B$. Therefore the direction $h = \mathbf{1}_A - \mathbf{1}_B$ preserves the total and harms no tight edge inequality. Every decreasing coordinate satisfies $x_v \geq \delta$ by nonnegativity of y_v. Finitely many slack constraints and positive decreasing coordinates ensure $x + \epsilon h \in C$ for all sufficiently small $\epsilon > 0$.

If $\min_B x > \min_A x$, a sufficiently small such step raises the lowest affected coordinates and decreases only strictly larger coordinates, strictly improving the ascending sorted vector. This contradicts full leximin optimality, including when smaller fixed coordinates occur elsewhere. Hence

$$\delta \leq \min_B x \leq \min_A x \leq x_a, \qquad y_a = x_a + \delta \leq 2x_a.$$

Interchange the two market sides for positive right-side changes. Nonpositive changes satisfy the claim immediately. Finally $\min(x_a, y_a) \geq y_a/2$, so the overlap is at least $T/2$ and the normalized distance is at most one. □

The constant two in the coordinatewise comparison is tight: on a single positive edge, leximin splits its value equally while a core endpoint gives its entire value to one player.

3 The sharp finite bound and its attainment

Proof of Theorem 1: upper bound. On matched pair i, write the change from x to z as $(d_i, -d_i)$, and put

$$q_i = |d_i|, \quad D = \sum_i q_i, \quad M = \max_a z_a > 0.$$

Thus $\|x - z\|_1 = 2D$. Both x payoffs in pair i are at least q_i: the decreasing endpoint by nonnegativity of z, and the increasing endpoint by Lemma 2. The increasing endpoint's z payoff is at least $2q_i$. Therefore

$$s_i \geq 2q_i, \qquad q_i \leq M/2, \qquad D \leq kM/2. \tag{2}$$

These statements also hold when $q_i = 0$.

Some player a satisfies $z_a = M$ and $x_a \geq M$. Otherwise, a sufficiently short move from z toward x lowers every maximum coordinate and leaves every other coordinate below M, contradicting minimax optimality. Since $M > 0$, this player belongs to a positive matched pair p. Its displacement is $q_p = x_a - M$; the other x payoff in that pair is at least q_p. Thus $s_p \geq M + 2q_p$. Summing this strengthened inequality with (2) for the other pairs gives

$$T \geq 2D + M \geq 2D\left(1 + \frac{1}{k}\right).$$

This proves the bound, also when $D = 0$.

If $k = 1$, all other payoffs are fixed at zero. The remaining two sum to T and range over a closed interval of splits. Maximizing their minimum is the same as minimizing their maximum, and uniquely selects the feasible split closest to equality. Hence $x = z$ in this case. □

Attainment and player count. For $k \geq 2$, take a complete $k \times k$ market with identical rows

$$(w_{i1}, \ldots, w_{ik}) = (1, \ldots, 1, 2).$$

Every perfect matching has value $T = k + 1$, and every edge belongs to one. Complementarity consequently makes every edge inequality tight throughout the core. Writing $c = (1, \ldots, 1, 2)$, the entire core is exactly

$$u_i = t \quad (1 \leq i \leq k), \qquad v_j = c_j - t \quad (1 \leq j \leq k), \qquad 0 \leq t \leq 1.$$

Its minimum coordinate is $\min(t, 1 - t)$, uniquely maximized at $t = 1/2$. Its maximum is $2 - t$, uniquely minimized at $t = 1$. Every coordinate changes by $1/2$, so $\|x - z\|_1 = k$ and $\rho = k/(k + 1)$. Adding one isolated player leaves the full selectors otherwise unchanged. Every matching on n players has at most $\lfloor n/2 \rfloor$ edges, which proves the exact n-player formula. □

4 Why exact assignment games have zero divergence

A game is exact when each coalition's worth is attained by some core allocation. The following known structural implication explains the boundary with unrestricted exact TU games; it is not claimed as new [3, 4].

Proposition 3. *If the matched-pair halfsplit h belongs to an assignment-game core, then* $\mathrm{Lmin}(C) = \mathrm{Lmax}(C) = h$. *Every exact assignment game satisfies this hypothesis.*

Proof. For any core allocation y, let P swap the endpoints of every fixed matched pair and fix unmatched coordinates. Then $h = (y + Py)/2$. Thus h is majorized by y: pairwise averaging can only increase sums of the smallest coordinates and decrease sums of the largest coordinates. If $y \neq h$, strict convexity of squared norm implies $\|h\|_2^2 < \|y\|_2^2$, so their sorted vectors cannot coincide. Therefore h strictly improves both relevant lexicographic orders.

For an exact assignment game, the two side coalitions L and R have worth zero and each attains that worth at a core allocation. Nonnegativity and matched-pair totals force these two allocations to put every s_i entirely on opposite sides. Their midpoint is $h \in C$. □

The attaining markets above are nonexact: their distinguished right player always receives at least one, despite having singleton worth zero. Thus the comparison is between two different restrictions of cooperative games: general assignment games have maximum divergence tending to one; within exact assignment games it is zero.

5 Evidence, related work, and review status

The argument above is a conventional all-parameter proof. Exact rational replays reconstruct all coalition values by enumerating matchings, verify all full selector stages for 255 nonzero 2×2 matrices with entries $0, 1, 2, 3$, and check the half-payoff guarantee by separately maximizing each coordinate over the entire core. They also exclude any violating minimax allocation, test rectangular and degenerate examples, and reconstruct attaining segment cores through $k = 6$. A separate implementation verifies ten even/odd attaining-family cases. Negative controls distinguish full leximin from first-stage maximin and minimax comparison points from arbitrary core allocations. The code and exact outputs accompany this note; finite tests do not replace its proof.

Sorted egalitarian selections and alternating-adjustment methods have older assignment-game precedents [2]; recent work develops fair-core algorithms [1]. The fair-price midpoint already provides a half-core-maximum benchmark for another solution concept [5]. Our candidate contribution is the guarantee specifically for full leximin and the sharp finite distance to minimax selections. A dated primary-source audit located no equivalent quantitative theorem, but does not certify novelty. The audit also records an apparent mismatch between the graph and printed vectors of Example 1 in Vazirani's v6, with exact reconstructed selectors and an independent public implementation fixture corroborating the reconstructed leximin. This discrepancy is separate from the proof above. External correctness, prior-art, and significance review remain pending.

References

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